

Chemistry

Unit 1 Review

The table below indicates the questions relevant to each study area:

Unit 1 Study Area	Section one	Section two
A. Properties and structure of atoms	1–12	1–10
B. Properties and structure of materials	13–31	11–26
C. Chemical reactions: reactants, properties and energy change	32–40	27–30

Section one—Multiple-choice questions

Question 1

Elements are placed into groups on the periodic table according to each element's:

- A. mass number.
- B. atomic radii.
- C. number of occupied electron shells.
- D. number of electrons in the outer shell.

Question 2

Which of the below correctly describes an anion that contains 11 protons, 12 neutrons and 10 electrons?

	Mass number	Charge
A.	11	–1
B.	11	+1
C.	23	–1
D.	23	+1

Question 3

Which one of the following elements has the highest electronegativity?

- A. B
- B. Be
- C. Ba
- D. Bi

Question 4

Which of the following pairs of elements are chemically most similar to each other?

- A. Li and Be
- B. Cl and Ar
- C. N and P
- D. Ne and Na

Question 5

Which one of the following metals reacts most readily with oxygen?

- A. Na
- B. K
- C. Mg
- D. Ca

Question 6

Group 18 elements are unreactive because:

- A. they have the largest atomic radii of all atoms in each period.
- B. they have a complete valence shell of electrons.
- C. they are gases.
- D. they can only form covalent bonds.

Question 7

Zinc is an element. Therefore:

- A. zinc has no isotope.
- B. all zinc atoms are identical.
- C. zinc atoms always contain the same number of protons.
- D. zinc atoms contain equal numbers of protons and neutrons.

Question 8

Copper has two isotopes, ^{63}Cu and ^{65}Cu , and a relative atomic mass of 63.55. Any one particular copper atom would:

- A. contain either 63 or 65 protons.
- B. contain either 34 or 36 neutrons.
- C. have a mass number of 63.55.
- D. have an atomic number of 29 or 31.

Question 9

Chlorine can exist as two isotopes, chlorine-35 and chlorine-37. Two samples of chlorine gas were prepared, one containing only chlorine-35 atoms and one containing only chlorine-37 atoms. Which one of the following statements about these samples is true?

- A. They would have different chemical properties.
- B. One mole of each sample would have the same mass.
- C. They would have different boiling points.
- D. They would have the same density.

Questions 10 and 11 refer to the following information.

The atomic number, mass number and electron configuration of four particles, W, X, Y and Z, are given in the following table.

Particle	Atomic number	Mass number	Electron configuration
W	17	37	2,8,7
X	19	39	2,8,8,1
Y	20	40	2,8,8
Z	19	40	2,8,8,1

Question 10

Which one of the following groups contains particles which are isotopes of each other?

- A. W and X only
- B. X and Z only
- C. Y and Z only
- D. W, X and Y only

Question 11

Which one of the following statements about particles W, X, Y and Z is correct?

- A. W is a noble gas.
- B. X is in group 4 of the periodic table.
- C. Y is a positively charged ion.
- D. Z is a negatively charged ion.

Question 12

An absorption spectrum is produced when electrons in an atom:

- A. release energy as they move from higher to lower energy levels.
- B. absorb energy as they move from higher to lower energy levels.
- C. release energy as they move from lower to higher energy levels.
- D. absorb energy as they move from lower to higher energy levels.

Question 13

Which one of the following methods of separation does not rely on the substances having different solubilities?

- A. crystallisation
- B. distillation
- C. filtration
- D. chromatography

Question 14

Which of the following is a heterogeneous mixture?

- A. carbon dioxide gas
- B. salt water
- C. brass
- D. orange juice

Question 15

Which of the following are pure substances?

- I. sodium chloride
- II. air
- III. dilute hydrochloric acid
- IV. magnesium metal

- A. I only
- B. I and II only
- C. IV and III only
- D. I and IV only

Question 16

Consider the following electrostatic forces.

- I. attraction between oppositely charged particles
- II. repulsion between positively charged particles
- III. repulsion between negatively charged particles

Which of these electrostatic forces are operating in an atom?

- A. I only
- B. I and II only
- C. I and III only
- D. I, II and III

Question 17

Magnesium is a group 2 element. Which of the following statements will **not** be true for magnesium?

- A. It will conduct electricity in the solid state.
- B. The formula of magnesium iodide will be MgI_2 .
- C. It will have the same number of valence electrons as calcium.
- D. Magnesium fluoride will conduct electricity in the solid state.

Question 18

Which one of the following formulae is **not** correct?

- A. AlPO_4
- B. CuSO_4
- C. $\text{Mg}_2(\text{PO}_4)_3$
- D. Na_2CO_3

Question 19

Element X is in group 4 of the periodic table. Element Y is in group 16. Which one of the formulae below is the most likely for a compound formed between X and Y?

- A. X_4Y_{16}
- B. X_{16}Y_4
- C. XY_4
- D. XY_2

Question 20

The electron configurations for four pairs of elements are shown below. Which pair would be most likely to form an ionic bond?

- A. (2,8,1) and (2,8,2)
- B. (2,8,1) and (2,8,7)
- C. (2,8,6) and (2,4)
- D. (2,8,7) and (2,8,6)

Question 21

Which one of the following statements about ionic bonding is not correct?

- A. When molten, ionic compounds are conductors of electricity.
- B. An ionic lattice contains both cations and anions in fixed positions.
- C. Ionic bonding involves the sharing of electrons between two different atoms.
- D. Compounds held together by ionic bonding generally have high melting temperatures.

Question 22

Which one of the following compounds will have the highest melting point?

- A. O₂
- B. CO₂
- C. Na₂O
- D. H₂O

Question 23

Which one of the following is **not** a characteristic of all ionic compounds?

- A. high melting point
- B. brittle
- C. conduct electricity in the solid state
- D. soluble in water

Question 24

Which one of the following describes a covalent bond?

- A. An electron shared between two atoms.
- B. Two electrons attracted to two atomic nuclei.
- C. A delocalised pair of electrons that can move between atoms.
- D. An attraction between two atoms with a significant difference in electronegativity.

Question 25

The formula for hydrogen cyanide is HCN. How many electrons are shared between the carbon and nitrogen atom?

- A. 2
- B. 4
- C. 6
- D. 8

Question 26

Which one of the following best explains why covalent molecular substances do not conduct an electrical current?

- A. They do not contain electrons.
- B. The valence shell of every atom is full.
- C. They contain particles that have no overall charge.
- D. They do not contain metals.

Question 27

Which one of the following is most likely to have the highest melting point?

- A. CO_2
- B. SO_2
- C. SiO_2
- D. C_4H_{10}

Question 28

Graphenes and fullerenes are classified as carbon nanomaterials because they contain nanoparticles. Substances that do not contain nanoparticles are called 'bulk materials'. Which of the statements below best explains why nanomaterials have properties different from bulk materials?

- A. Nanoparticles are smaller than molecules so they are unstable.
- B. Nanoparticles are electrically charged.
- C. Nanoparticles have a larger relative surface area compared to particles in bulk materials.
- D. Nanoparticles contain electrons in excited states which makes them more reactive.

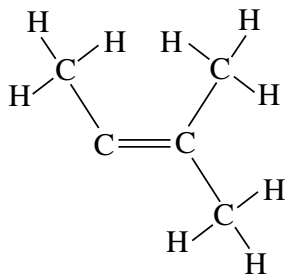
Question 29

Which one of the following lists only contains alkanes?

- A. CH_4 C_4H_{10} C_8H_{16}
- B. C_2H_6 C_4H_{10} C_8H_{18}
- C. C_2H_6 CH_4 $\text{C}_2\text{H}_5\text{OH}$
- D. C_2H_4 C_4H_8 C_6H_{12}

Question 30

Consider the following molecule:



Which one of the following is the IUPAC name for this hydrocarbon?

- A. 2-methylbut-2-ene
- B. 1,2,2-trimethylethene
- C. 3-methylbut-2-ene
- D. 3,3-dimethylprop-2-ene

Question 31

Which one of the following will **not** undergo addition reactions?

- A. pent-2-ene
- B. methylbutane
- C. pent-1-ene
- D. 3-methylbut-1-ene

Question 32

Which of the following is the correct equation for the **complete** combustion of octane?

- A. $2\text{C}_8\text{H}_{18} + 9\text{O}_2 \rightarrow 16\text{C} + 18\text{H}_2\text{O}$
- B. $\text{C}_8\text{H}_{18} + 25\text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$
- C. $2\text{C}_8\text{H}_{18} + 25\text{O}_2 \rightarrow 16\text{CO}_2 + 18\text{H}_2\text{O}$
- D. $2\text{C}_8\text{H}_{18} + 17\text{O}_2 \rightarrow 16\text{CO} + 18\text{H}_2\text{O}$

Question 33

It was found that 0.235 mol of a hydrocarbon has a mass of 16.0 g. Which one of the following is the hydrocarbon?

- A. CH_4
- B. C_3H_6
- C. C_5H_8
- D. C_6H_{14}

Question 34

The total number of atoms in 4 moles of pentane is:

- A. 68
- B. 4.1×10^{25}
- C. 2.4×10^{24}
- D. 3.6×10^{25}

Question 35

Which of the following is the percentage by mass of hydrogen in water?

- A. 89%
- B. 22%
- C. 11%
- D. 5.5%

Question 36

A particular brand of plant fertiliser contains 58.5% urea, $\text{CO}(\text{NH}_2)_2$, as the only nitrogen containing compound. The mass of nitrogen in 125 g of this fertiliser is:

- A. 29.2 g
- B. 40.0 g
- C. 50.0 g
- D. 73.1 g

Question 37

Which of the following is the mass of 2 moles of carbon dioxide?

- A. 88 g
- B. 44 g
- C. 24 g
- D. 12×10^{23} g

Question 38

Which one of the following contains the same number of ions as 1 mole of NaOH?

- A. 1 mole of Na_2S
- B. 0.5 mole of K_3PO_4
- C. 1 mole of MgCl_2
- D. 1.5 moles of MgO

Question 39

Which of the following processes is exothermic?

- A. ice melting
- B. chemical bonds breaking
- C. evaporation of petrol
- D. crystallisation

Question 40

When sodium metal reacts with water, a violent exothermic reaction occurs. Which one of the following statements is correct?

- A. The temperature of the water will decrease.
- B. The enthalpy of the products is less than the enthalpy of the reactants.
- C. Heat energy is the only form of energy released to the surroundings.
- D. ΔH is positive for the process.

Section two—Short and extended answer questions

Question 1

- Describe how Bohr used evidence from spectral analysis to propose his model of the atom.
- Give two specific limitations of Bohr's model of the atom.

Question 2

Iridium has two naturally occurring isotopes. Their relative abundances and masses are shown in the following table.

Isotope	Relative isotopic mass	Percentage abundance
^{191}Ir	190.97	37.30%
^{193}Ir	192.97	62.70%

- What is the name of the instrument that is commonly used to experimentally obtain the abundances and relative isotopic masses data?
- Using the information in the table, determine the relative atomic mass of iridium.

Question 3

Isotopes of an element have the same chemical properties but different physical properties. Explain why this is.

Question 4

The following processes occur in a mass spectrometer.

- Acceleration
 - Deflection
 - Detection
 - Ionisation
 - Vaporisation
- List these processes in the order in which they occur.
 - Briefly describe each process.

Question 5

Give a concise explanation for each of the following.

- The atomic radius of chlorine is smaller than that of sodium.
- The first ionisation energy of fluorine is higher than that of lithium.
- The reactivity of Be is less than that of Ba.

Question 6

Answer the questions below using only the following elements. For each answer you need to use your knowledge of atomic structure to explain your choice(s).

fluorine calcium potassium aluminium
carbon nitrogen neon magnesium

- Which two of these will have similar chemical properties? Explain your answer.
- Which one of these will be unreactive? Explain your answer.
- Which one of these will have the lowest first ionisation energy? Explain your answer.
- What will be the formula of the compound formed between aluminium and sulfur? Explain your answer.

Question 7

- a. Use the periodic table to write a correct symbol for each of the following.
 - i. the element that is in group 2 and period 4
 - ii. an inert gas with exactly three occupied electron shells
 - iii. an element from group 14 that is a non-metal
 - iv. an alkali metal that has only three occupied electron shells
 - v. the element in period 2 that has the largest atomic radius
 - vi. the element in group 15 that has the highest first ionisation energy
 - vii. the element in period 2 with the highest electronegativity
- b. Describe the trend in chemical reactivity of elements going down group 1 of the periodic table and give a brief explanation for this trend.

Question 8

Give brief definitions of the following terms.

- a. mass number
- b. atomic number
- c. relative isotopic mass
- d. Avogadro's number
- e. ion

Question 9

In atomic absorption spectroscopy (AAS), radiation of a specific wavelength is passed through a vaporised and ionised sample of the material. Some of the atoms will absorb this radiation.

- a. Explain why the wavelength of the radiation has to be specific to the element being analysed.
- b. Flame tests are an example of atomic emission. Explain how the process of atomic emission is different from atomic absorption.

Question 10

Explain what is meant by a homogeneous mixture. Give an example of a homogeneous mixture from the home to illustrate your answer.

Question 11

Write the formula of each of the following compounds.

- a. potassium sulfate
- b. aluminium oxide
- c. sodium nitrate
- d. iron(III) sulfide

Question 12

Complete the following table by writing the name of each of the compounds.

	Formula	Name
a.	$\text{Ca}(\text{NO}_3)_2$	
b.	CuCO_3	
c.	$\text{Mg}(\text{OH})_2$	
d.	Cr_2O_3	

Question 13

Draw the structure of the organic compounds produced in the following reactions.

- a. aqueous bromine (bromine water) + propene
- b. chlorine gas + ethane (in the presence of ultraviolet light)
- c. hydrogen + 2-methylbut-2-ene

Question 14

Magnesium is a commonly used structural metal.

- a. Write the electronic configuration of a magnesium atom.
- b.
 - i. The radius of a magnesium atom is 160 pm ($1\text{ pm} = 10^{-12}\text{ m}$). What is the radius of the magnesium atom in nanometres?
 - ii. Would you predict the radius of a sodium atom to be smaller or larger than 160 pm? Explain your answer.
- c.
 - i. Describe the model commonly used to describe the structure of metals such as magnesium and the nature of the bonding between its particles. You may include a labelled diagram in your answer.
 - ii. Use the metallic bonding model to explain why magnesium is a good conductor of electricity.
 - iii. The metallic bonding model is useful to explain many of the properties of metals. However, like most models, it has some limitations. Give one example of the limitations of the metallic bonding model.
- d.
 - i. Magnesium reacts readily with dilute hydrochloric acid. What would you observe if a small piece of magnesium was added to some hydrochloric acid? Write a chemical equation for this reaction.
 - ii. Give the symbol for one metal that would be expected to react more vigorously with hydrochloric acid than magnesium does.
- e. Magnesium is most commonly used as an alloy for structural purposes. What is an alloy? Give one example of how alloying affects the properties of a metal.

Question 15

Ionic compounds have a variety of uses in everyday life.

- a. Give the formula, and a use, of one ionic compound.
- b. Use the ionic bonding model to explain the following properties of ionic compounds.
 - i. Solid ionic compounds cannot conduct electricity.
 - ii. Ionic compounds are generally brittle.
 - iii. Ionic compounds have high melting temperatures.

Question 16

Draw electron dot diagrams to show the bonding present in the following ionic compounds.

- a. NaCl
- b. MgBr_2

Question 17

When magnesium burns in oxygen, a very bright white flame is produced and the white solid, magnesium oxide, is formed.

- a. Describe the bonding in:
 - i. magnesium.
 - ii. magnesium oxide.
- b. Use your answer to part (a) to explain:
 - i. why magnesium metal can conduct electricity in the solid state and magnesium oxide cannot.
 - ii. why magnesium oxide is brittle and magnesium is malleable.

Question 18

This question is about the metal aluminium.

- a. Explain why aluminium is less reactive than sodium.
- b. Draw a labelled diagram to show the bonding present in pure aluminium.
- c. Explain why the atomic mass of aluminium is not a whole number.

Question 19

Draw electron dot diagrams for the following molecules.

- a. H_2
- b. O_2
- c. NH_3
- d. H_2O

Question 20

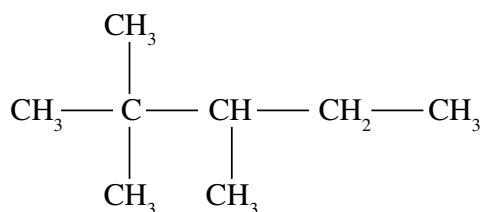
Diamond and graphite are allotropes of carbon. Although they are similar in some respects, they are very different in their structure and uses.

- a. Give a brief definition of the term 'allotrope'.
- b. Describe the similarity in the bonding of these two materials.
- c. Compare the structures of diamond and graphite.
- d. Graphite is an excellent conductor of electricity while diamond is unable to conduct electricity. Explain this difference in terms of the structure of the lattices.
- e. Graphite is used as a lead in pencils. Explain why graphite can be used this way, with reference to the bonding in the lattice.

Question 21

Give the name of each of the following compounds.

- a. $\text{CH}_2=\text{CHCH}_2\text{CH}_3$
- b. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- c.



Question 22

Give the displayed structural formula of each of the following compounds.

- 3-ethylpentane
- hept-1-ene
- 2,3-dimethylbut-2-ene
- cis but-2-ene

Question 23

When hydrocarbons burn in air they react with the oxygen in the air. In excess oxygen they produce carbon dioxide and water. They can also react to produce carbon monoxide and water, and even just carbon and water.

- Write balanced chemical equations for the three possible combustion reactions of propane (C_3H_8).
- Use the equations to comment on the amount of oxygen gas required for each of the three reactions.

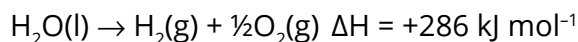
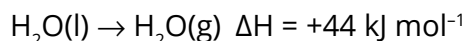
Question 24

Crude oil (petroleum) is an important resource to our society.

- What is the origin of crude oil?
- Crude oil is a mixture.
 - Briefly describe the composition of crude oil.
 - Give the molecular formula of two hydrocarbons present in crude oil.

Question 25

The enthalpy changes for two processes are shown below.



- Explain why both values are positive.
- Explain the large difference between the two values.

Question 26

A sample of aluminium nitrate, $\text{Al}(\text{NO}_3)_3$, has a mass of 30.5 g.

- Calculate the number of moles of aluminium nitrate in the sample.
- Calculate the number of moles of nitrogen atoms present.
- Determine the percentage by mass of oxygen in aluminium nitrate.

Question 27

Element X forms a compound with oxygen in which the ratio of the number of atoms of X to the number of atoms of oxygen is 2:3. X makes up 59.7% of the mass of the compound.

Calculate the molar mass of X and use it to identify this element.

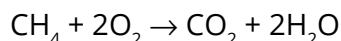
Question 28

Potassium nitrate, KNO_3 , and ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, can both be used in fertilisers as a source of nitrogen for plants.

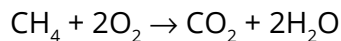
What mass of ammonium sulfate contains the same amount of nitrogen as 65.0 g of potassium nitrate?

Question 29

- a. Use the following equation to calculate how many moles of oxygen would be needed to react with 1.34 moles of methane (CH_4).



- b. Use the following equation to calculate the mass of water that would be produced when 2.50×10^{-2} moles of methane (CH_4) is burned in excess oxygen.



Question 30

Examine the data for two fuels, butane and ethanol, shown in the following table.

	Butane (C_4H_{10})	Ethanol ($\text{C}_2\text{H}_5\text{OH}$)
Boiling point	-0.5°C	78.4°C
Equation for combustion	$2\text{C}_4\text{H}_{10} + 13\text{O}_2 \rightarrow 8\text{CO}_2 + 10\text{H}_2\text{O}$	$\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$
Enthalpy (heat) of combustion	2859 kJ mol^{-1}	1367 kJ mol^{-1}

- a. Which of the fuels will be a gas at room temperature?
- b. Calculate the number of moles of butane in one 220 g canister.
- c. Calculate the maximum amount of energy that could be released from 220 g of butane.
- d. Explain why it is unlikely that this amount of energy would be released.
- e. Calculate the mass of oxygen required for the combustion of 220 g of butane.
- f. Compare the carbon emissions of the two fuels by calculating the number of moles of carbon dioxide given off by 100 g of each of the fuels when they combust.
- g. Explain why the amount of carbon emissions is important when considering the use of fuels.